MATH 241: Calculus III

Fall 2026

Lecture 2: Vectors in $\mathbb{R}^3$

Background & Motivation

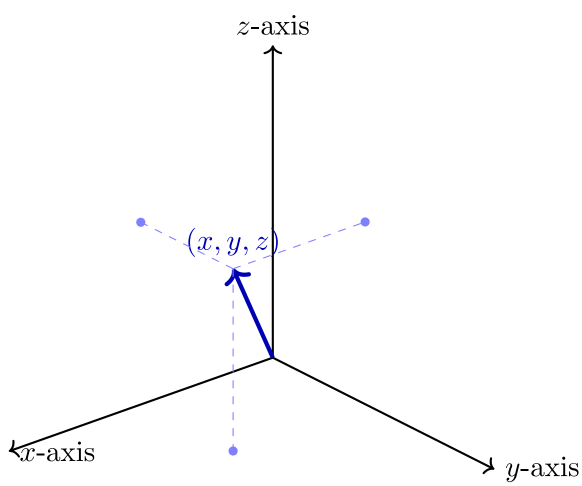
Last time we set up 3D coordinates. Now we need a language to describe direction and magnitude in space—that's what vectors give us. Vectors are the building blocks of everything in this course: lines, planes, curves, derivatives, integrals, and fields all use vector notation.

1. Vectors — Definitions

Definition 1: Vector

A **vector** is a quantity carrying both __________ and __________ — visually, an arrow. Two arrows are considered the *same* vector whenever they have the same length and direction, no matter where they are drawn, so a vector is really an equivalence class of parallel arrows. Placing its tail at the origin, the tip lands at a specific point (v_1, v_2, v_3), and those coordinates are the **components** of $\mathbf{v}$.

– Written $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$; angle brackets distinguish vectors from points (a, b, c)



A vector in $\mathbb{R}^3$ represented by its components along the x, y, and z axes. Dashed lines mark the projections onto the three coordinate planes.

Definition 2: Magnitude (Norm)

The **magnitude** (or **norm**) $\|\mathbf{v}\|$ is the length of the vector — the Euclidean distance from its tail to its tip. When $\mathbf{v}$ is drawn with its tail at the origin, $\|\mathbf{v}\|$ is literally the distance from the origin to the terminal point, and it is unchanged by rotating or translating the coordinate system. In short, magnitude strips away direction and records only *how much* of the vector there is. It is computed by the Pythagorean theorem applied to the components:

$$\|\mathbf{v}\| = \underline{\hspace{6cm}}$$

Definition 3: Unit Vector

A vector with magnitude _____. To **normalize**:

$$\hat{\mathbf{v}} = \underline{\hspace{6cm}}$$

Standard unit vectors: $\hat{\mathbf{i}} = \langle 1, 0, 0 \rangle$, $\hat{\mathbf{j}} = \langle 0, 1, 0 \rangle$, $\hat{\mathbf{k}} = \langle 0, 0, 1 \rangle$

Definition 4: Direction Cosines

For $\mathbf{v} = \langle a, b, c \rangle$, the angles α, β, γ with the x-, y-, z-axes:

$$\cos\alpha = \underline{\hspace{3cm}}, \quad \cos\beta = \underline{\hspace{3cm}}, \quad \cos\gamma = \underline{\hspace{3cm}}$$

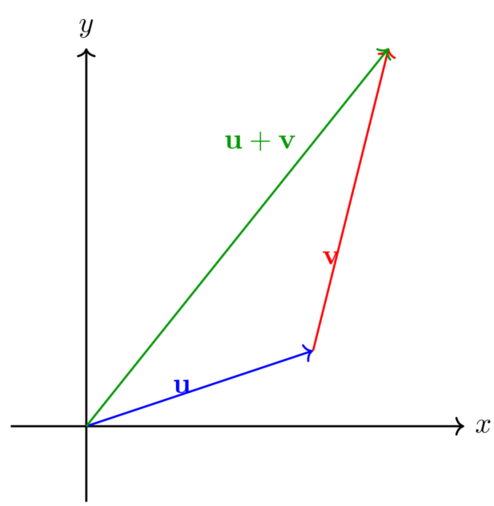
Key identity: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \underline{\hspace{3cm}}$

Vector Arithmetic

If $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$:

$$\mathbf{u} + \mathbf{v} = \underline{\hspace{6cm}}$$
$$c\,\mathbf{u} = \underline{\hspace{6cm}}$$

– Addition is "tip-to-tail"; scalar multiplication scales length



Tip-to-tail method of vector addition: placing $\mathbf{v}$ at the tip of $\mathbf{u}$ results in $\mathbf{u} + \mathbf{v}$.

2. The Dot Product

Definition 5: Dot Product

Component formula:

$$\mathbf{u} \cdot \mathbf{v} = \underline{\hspace{12cm}}$$

Geometric formula:

$$\mathbf{u} \cdot \mathbf{v} = \underline{\hspace{12cm}}$$

– The dot product is a ________ (number, not a vector)

Dot Product Facts

- $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} =$ ________________
- Angle between vectors: $\theta =$ ________________

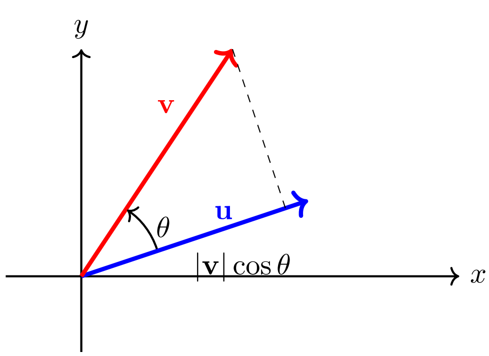
Definition 6: Vector Projection

The **projection** of $\mathbf{u}$ onto $\mathbf{v}$ is the "shadow" that $\mathbf{u}$ casts along the line of $\mathbf{v}$ — the part of $\mathbf{u}$ that points in the direction of $\mathbf{v}$. It is a vector parallel to $\mathbf{v}$ whose length is the *scalar component* $\text{comp}_{\mathbf{v}}\, \mathbf{u} = \|\mathbf{u}\| \cos\theta$.

$$\text{proj}_{\mathbf{v}}\, \mathbf{u} = \underline{\hspace{12cm}}$$

Scalar component:

$$\text{comp}_{\mathbf{v}}\, \mathbf{u} = \underline{\hspace{12cm}}$$



The dot product interpreted as the projection of $\mathbf{v}$ onto $\mathbf{u}$, scaled by $\cos\theta$.

Example 1: Crane Cable Forces

Two crane cables exert forces $\mathbf{F}_1 = \langle 120, 0, 80 \rangle$ N and $\mathbf{F}_2 = \langle -50, 75, 60 \rangle$ N.

(a) Find the total force $\mathbf{F}_1 + \mathbf{F}_2$.

(b) Find a unit vector in the direction of the total force.

(c) What force $\mathbf{F}_3$ would perfectly counterbalance $\mathbf{F}_1 + \mathbf{F}_2$?

Example 2: Dot Product and Angle

Let $\mathbf{a} = \langle 3, 0, 4 \rangle$ and $\mathbf{b} = \langle 1, 2, -1 \rangle$. Find $\mathbf{a} \cdot \mathbf{b}$ and the angle between them.

Example 3: Projection

Project $\mathbf{a} = \langle 3, 4, 5 \rangle$ onto $\mathbf{n} = \langle 1, 1, 1 \rangle$.

3. The Cross Product

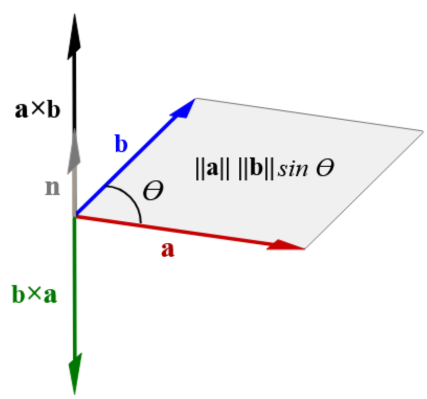
Definition 7: Cross Product

Determinant formula:

$$\mathbf{u} \times \mathbf{v} = \underline{\hspace{6cm}}$$

Properties:

- $\mathbf{u} \times \mathbf{v}$ is $\underline{\hspace{6cm}}$
- $\|\mathbf{u} \times \mathbf{v}\| = \underline{\hspace{3cm}}$ – area of parallelogram
- Direction via the $\underline{\hspace{3cm}}$
- $\mathbf{u} \times \mathbf{v} = \underline{\hspace{3cm}}$ (anti-commutative)



Cross product: $\mathbf{a} \times \mathbf{b}$ is perpendicular to both, with magnitude $= \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$ (parallelogram area).

Cross Product Facts

- $\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \underline{\hspace{3cm}}$
- $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$
- $\mathbf{u} \times \mathbf{u} = \mathbf{0}$ always

Example 4: Cross Product Computation

Compute $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 3, 0, 4 \rangle$ and $\mathbf{b} = \langle 1, 2, -1 \rangle$.

4. Scalar Triple Product

Scalar Triple Product

$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ – gives the ____________________ spanned by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

$$\text{Volume} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

Example 5: Scalar Triple Product — Volume

Find the volume of the parallelepiped spanned by $\mathbf{a} = \langle 1, 0, 2 \rangle$, $\mathbf{b} = \langle 3, 1, 0 \rangle$, $\mathbf{c} = \langle 0, 2, 1 \rangle$.

Show all work for full credit.

Practice Problems

Problem 1. Let $\mathbf{v} = \langle 3, 4, 12 \rangle$. Find $\|\mathbf{v}\|$, the unit vector $\hat{\mathbf{v}}$, and the direction cosines.

Problem 2. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 0, -2 \rangle$.

Problem 3. Find the angle between $\mathbf{u} = \langle 1, 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 1 \rangle$.

Problem 4. Find $\text{proj}_{\mathbf{n}}\, \mathbf{a}$ where $\mathbf{a} = \langle 3, 4, 5 \rangle$ and $\mathbf{n} = \langle 1, 1, 1 \rangle$. Also find the perpendicular component.

Problem 5. Three points $P_1 = (1, 2, 3)$, $P_2 = (4, -1, 2)$, $P_3 = (-2, 3, 5)$ define a plane. Use the cross product to find a normal vector.

Problem 6. Find the volume of the parallelepiped spanned by $\mathbf{a} = \langle 2, 1, 0 \rangle$, $\mathbf{b} = \langle 1, -1, 3 \rangle$, $\mathbf{c} = \langle 0, 2, 1 \rangle$.